

Cyber Tabletop Exercise

Food Processing Ransomware Scenario

Prepared for: Food Processing Cooperative
Scenario source: STIX 2.1 bundle supplied by the user
Facilitation-ready exercise pack | Confidential – Exercise Use Only

Scenario premise

Ransomware disrupts scheduling, inventory visibility, and cold chain monitoring during active production.

Exercise duration	90 minutes, adaptable to 60 minutes
Audience	Executive crisis cell, plant operations, IT/SOC, food safety/quality, legal, communications, supply chain
Threat actor	Ransomware Operator (crime-syndicate; motivation: financial-gain)
Malware	Enterprise Ransomware Payload
Sector	agriculture, food-production

1. Exercise Overview

This tabletop exercise simulates a financially motivated ransomware incident affecting a food processing cooperative during active production. The exercise is designed to test crisis coordination across cyber response, plant operations, food safety, supply chain, customer communications, executive decision-making, and recovery governance.

Exercise assumptions

- The scenario is fictional but should be treated as real during discussion.
- The cooperative operates production scheduling, inventory visibility, warehouse management, and cold chain monitoring platforms across corporate IT and plant operations environments.
- The ransomware primarily affects IT-hosted business and production-support systems; no unsafe manipulation of physical processing equipment is assumed unless players identify evidence during the exercise.
- Food safety decisions must be made using verified temperature logs, manual checks, quality assurance procedures, and documented risk acceptance.
- SOC containment and recovery decisions remain technical-accountable decisions; the crisis cell sets business priorities, approves risk decisions, and coordinates internal and external communications.

Exercise objectives

- 1 Validate escalation from IT/SOC and plant operations to the crisis management structure.
- 2 Assess the organization's ability to maintain safe production when scheduling, file repositories, inventory visibility, and cold chain dashboards are degraded.
- 3 Clarify decision rights for shipment holds, product release, production continuation, regulatory notification, and customer communication.
- 4 Test the practicality of manual workarounds for production scheduling, inventory reconciliation, temperature validation, and order fulfillment.
- 5 Evaluate ransomware response fundamentals: containment, evidence preservation, backup validation, restoration prioritization, ransom decision governance, and post-incident review.

Rules of engagement

- Real incidents and safety issues take priority over the exercise.
- There are no right or wrong answers; the goal is to identify strengths, gaps, and decision bottlenecks.
- Participants should answer according to their actual roles, procedures, and available information.
- Do not fight the scenario. Ask clarifying questions, but make decisions with imperfect information.
- Facilitators may accelerate or slow inject timing to support learning objectives.

Suggested participants

Role	Primary exercise focus
Executive / Crisis Lead	Set priorities, approve major business decisions, chair crisis rhythm.
Plant Manager / Operations	Assess production impact, manual scheduling, shift safety, throughput, staffing.
SOC / IT Incident Response	Triage, containment, forensics, eradication, restoration, reconnection decisions.
Quality & Food Safety	Evaluate cold chain evidence, product hold/release, food safety risk decisions.

Supply Chain / Logistics	Manage inbound materials, outbound shipments, customer delivery commitments.
Legal / Privacy / Compliance	Advise on reporting, preservation, law enforcement, contractual obligations.
Communications / Customer Relations	Prepare internal, supplier, customer, and media messaging.
Finance / Insurance	Track financial impact, insurance notification, ransom/payment governance.

2. Scenario Background

The Food Processing Cooperative is operating during a high-volume production period. Production planners rely on scheduling applications and shared repositories for shift plans, production orders, sanitation schedules, and product specifications. Warehouse teams depend on inventory visibility to match finished goods, raw materials, and customer orders. Quality and food safety teams rely on cold chain monitoring dashboards for refrigerated storage, in-process holds, and release decisions.

At the start of the exercise, several production-support systems become unavailable. The first signs appear operational rather than security-related. As the incident develops, ransomware indicators are identified and the organization must decide how to contain the cyber incident while protecting food safety, customers, suppliers, and business continuity.

In-scope systems and processes

- Production scheduling system and shared file repositories.
- Inventory visibility, warehouse status, and shipment preparation processes.
- Cold chain monitoring dashboards and supporting data feeds.
- Manual temperature checks, product hold/release process, and quality assurance records.
- SOC triage, endpoint/server containment, backup validation, and phased restoration.

Out-of-scope assumptions

- No confirmed threat to human safety or physical processing control systems at the start of the exercise.
- No confirmed customer or employee personal data exfiltration unless introduced by the facilitator as an optional inject.
- No ransom payment decision is forced; the discussion should focus on governance, legal constraints, and business impact.

Success criteria

- Clear incident commander and crisis lead are identified within the first inject.
- Critical production and food safety decisions are documented with accountable owners.
- Manual workarounds are prioritized according to safety, regulatory, customer, and financial impact.
- Containment and restoration decisions are sequenced to avoid reinfection and preserve evidence.
- After-action gaps are translated into owners, timelines, and measurable improvements.

3. Exercise Timeline and Injects

Exercise time	Event	Description
09:00	Inject 1	Scheduling system outage and shared repository unavailability during active production shift.
10:15	Inject 2	Cold chain dashboards stop updating; food safety and product release uncertainty emerges.
12:00	Inject 3	Plant management considers pausing outbound shipments pending product integrity validation.
14:30	Inject 4	Customers and suppliers request delivery assurance and confirmation that food safety controls remain effective.
Next business day	Debrief	Recovery priorities, post-incident reporting, lessons learned, and control improvements.

Inject 1 – Scheduling System Outage

09:00 – Active production shift

STIX source inject: Production scheduling systems and shared file repositories become unavailable during an active production shift.

Line supervisors report that production schedules are unavailable and shared file repositories are not accessible. Some operators are using printed schedules from the prior shift, while others are waiting for updated production orders. IT initially treats the event as a system availability incident. Within 20 minutes, multiple users report unusual file extensions and ransom-note text appearing in shared folders.

What we know

- Scheduling system unavailable for production planners and supervisors.
- Shared repositories cannot be opened from several plant and office workstations.
- No confirmed impact to processing machinery or safety systems at this stage.
- An incident ticket has been opened and the SOC is engaged for triage.

Discussion questions

- Who declares this a cyber incident versus a major IT incident, and what is the escalation path?
- Which production activities can continue safely without the scheduling system?
- What manual workarounds are authorized, and who validates that they use the latest production specifications?
- What evidence should be preserved before systems are reimaged or restored?
- Who communicates to plant staff, and what should they be told not to do?

Expected actions / good-practice indicators

- Assign incident commander, crisis lead, scribe, and plant operations liaison.
- Freeze affected shared repositories and preserve logs, endpoint telemetry, ransom notes, and timestamps.
- Confirm whether backups are available and isolated before restoration attempts.
- Use only approved manual schedules/specifications and document deviations.
- Initiate a crisis rhythm with decision log and stakeholder map.

Facilitator note: Look for confusion between SOC containment authority and crisis cell business decision authority. Probe whether participants know where approved offline production specifications are stored.

Inject 2 – Cold Chain Monitoring Loss

10:15 – 75 minutes after discovery

STIX source inject: Cold chain dashboards stop updating, creating uncertainty about food safety and product release decisions.

Quality assurance reports that cold chain dashboards have stopped updating for refrigerated storage areas and several in-transit staging zones. Temperature sensors may still be recording locally, but central visibility is unavailable. Operators are unsure whether current product lots can remain in production, be moved to hold, or be released.

What we know

- Cold chain dashboards are stale; last central update was approximately 45 minutes ago.
- Manual thermometer checks are available but not yet consistently recorded.
- Several high-value perishable lots are scheduled for packaging and outbound shipment today.
- SOC has identified ransomware-like encryption on multiple file servers and is isolating affected segments.

Discussion questions

- What minimum evidence is required to release, hold, or discard product?
- Who has authority to stop production or place product on quality hold?
- How are manual temperature checks assigned, documented, reviewed, and retained?
- How should IT containment actions be coordinated with operations to avoid unintended food safety impact?
- What are the triggers for notifying regulators, customers, insurers, or law enforcement?

Expected actions / good-practice indicators

- Quality/Food Safety lead establishes product disposition criteria and manual monitoring cadence.
- Operations identifies critical lots, shelf-life constraints, and storage capacity limitations.
- SOC prioritizes systems that support cold chain evidence and prevents premature reconnection.
- Legal/compliance reviews reporting thresholds and contractual notification obligations.
- Crisis cell documents risk decisions and rationale for any continued production.

Facilitator note: Push participants to separate availability of dashboards from actual temperature control. Ask how they prove product integrity if central telemetry is unavailable.

Inject 3 – Shipment Pause Decision

12:00 – Midday production and loading window

STIX source inject: Plant management considers pausing outbound shipments until product integrity can be validated.

Distribution is ready to load outbound shipments, but plant management is considering pausing shipments until product integrity can be validated. Sales teams warn that key customers have delivery windows with penalties. Warehouse staff cannot fully reconcile inventory status because visibility tools remain degraded.

What we know

- Outbound shipments include refrigerated products with time-sensitive delivery commitments.
- Inventory reports are incomplete; some lot-location records are being maintained manually.
- Backups exist, but IT has not confirmed whether they are clean or current enough for restoration.
- No evidence of exfiltration has been confirmed, but ransomware notes threaten public disclosure if payment is not made.

Discussion questions

- What decision criteria determine whether shipments continue, pause, or partially resume?
- Who approves customer-specific exceptions or priority shipments?
- How are lot traceability, chain of custody, and temperature records preserved during manual operations?
- What is the organization's ransom engagement decision process, including legal and insurance input?
- How are executives briefed on operational, safety, regulatory, and financial tradeoffs?

Expected actions / good-practice indicators

- Implement shipment hold/release matrix based on product risk, evidence quality, customer priority, and regulatory exposure.
- Prioritize restoration of inventory, quality records, and scheduling based on business impact analysis.
- Establish customer messaging that is accurate, limited, and aligned across sales, legal, and communications.
- Confirm backup integrity in a clean environment before restoring production-support systems.
- Do not negotiate or pay ransom without executive, legal, insurance, and law-enforcement-informed governance.

Facilitator note: Ask participants to make an explicit decision: full pause, partial pause, or continue with controls. Capture decision owner, evidence, risk acceptance, and communication plan.

Inject 4 – Supplier and Customer Pressure

14:30 – External pressure increases

STIX source inject: Customers and suppliers request delivery assurance and confirmation that food safety controls remain effective.

Strategic customers request delivery assurance and confirmation that food safety controls remain effective. Suppliers ask whether inbound deliveries should continue. A local industry contact reports rumors of a ransomware incident affecting the cooperative. Communications teams ask whether to prepare a holding statement.

What we know

- Customers are seeking confirmation of delivery status and food safety controls.
- Suppliers need instructions on inbound deliveries and staging availability.
- SOC continues containment and forensics; initial restoration may take 24–48 hours for key business systems.
- Plant operations can continue only with reduced throughput and manual quality assurance controls.

Discussion questions

- What can be said externally without overcommitting or creating legal exposure?
- Which customers, suppliers, regulators, and industry partners require proactive outreach?
- How will the organization coordinate messages across operations, sales, legal, communications, and executives?
- What metrics should be tracked for operational impact, food safety, financial loss, and recovery progress?
- What must be included in the post-incident report and after-action improvement plan?

Expected actions / good-practice indicators

- Approve a single source of truth for stakeholder messaging and a cadence for updates.
- Create supplier/customer segmentation and prioritize communications based on safety, contractual, and revenue impact.
- Maintain incident log, decision log, communications log, and product disposition evidence.
- Define restoration criteria: clean backups, endpoint/server validation, credential resets, monitoring, and business acceptance testing.
- Schedule hot wash and assign preliminary remediation owners before the crisis cell stands down.

Facilitator note: Listen for premature statements such as 'no food safety impact' before evidence exists. Encourage phrases that distinguish known facts, actions underway, and next update timing.

4. Facilitator Guide

Opening script

Today's exercise is a discussion-based tabletop. The scenario is fictional, but please respond as you would in a real incident. We will introduce limited information through injects. Your task is to identify actions, decision owners, communications, and gaps. The exercise is not a test of individuals; it is a test of coordination, procedures, and assumptions.

Facilitation prompts by audience

Audience	Prompt focus
Executive	What decisions require executive approval? What is the current risk posture? What tradeoffs are acceptable between safety, customer delivery, and financial loss?
Operational	Can production continue safely? Which lots, lines, warehouses, and shipments are most constrained? What manual processes are realistic for several shifts?
Technical	What is affected, what is contained, what evidence is preserved, what is the restore order, and what criteria make reconnection safe?
Legal / Compliance	What reporting thresholds, contractual notifications, litigation holds, insurance notices, and law-enforcement considerations apply?
Communications	What facts are approved, what cannot be said yet, who approves messages, and when will stakeholders receive the next update?

Decision points to capture

- Incident severity and escalation level.
- Whether to continue, reduce, or pause production.
- Whether to hold, release, or discard affected product lots.
- Whether to pause outbound shipments, partially resume, or continue under manual controls.
- Restoration priority and reconnection acceptance criteria.
- External notification and customer/supplier messaging approach.
- Ransomware governance position and engagement constraints.

Artifacts to request during play

- Current incident response plan and ransomware playbook.
- Crisis management structure, escalation matrix, and contact lists.
- Business continuity plan for production scheduling and quality records.
- Backup inventory and restore test evidence for critical systems.
- Food safety product hold/release procedure and cold chain monitoring SOP.
- Customer and supplier communication templates.
- Regulatory notification decision tree and post-incident report template.

5. Debrief and Lessons Learned

Hot wash questions

- What decisions were made quickly and confidently?
- Where did participants lack information, authority, or clear ownership?
- Which manual workarounds were realistic for one shift, and which would fail over 24–48 hours?
- What evidence was required to support food safety, shipment release, and stakeholder communications?
- Which technical dependencies should be restored first, and were those priorities already documented?
- What should be changed in playbooks, contact lists, supplier/customer communications, backup testing, or crisis governance?

Observation scoring

Observation area	1 - Low confidence	3 - Moderate confidence	5 - High confidence
Escalation and governance	Roles unclear; escalation delayed	Roles identified but decision rights debated	Incident/crisis roles and decision rights clear
Food safety continuity	Manual controls unclear or undocumented	Manual checks possible with gaps	Manual monitoring, hold/release, and evidence process understood
Technical containment	Containment ad hoc; evidence risk	SOC actions defined but dependencies unclear	Containment, forensics, backup validation, restoration sequencing clear
Business continuity	Workarounds informal	Workarounds known but untested	Manual processes tested, documented, staffed, and time-bounded
Communications	Inconsistent or premature messaging	Basic holding lines identified	Approved messaging, owners, cadence, and stakeholder segmentation clear

Improvement plan template

Finding	Risk	Owner	Target date	Success measure

6. MITRE ATT&CK Mapping

The STIX bundle explicitly describes ransomware disruption but does not include detailed initial access or execution objects. The mapping below therefore separates scenario-confirmed behavior from plausible ransomware behaviors used to support discussion and detection planning.

Tactic	Technique	Use in scenario	Basis
Initial Access	T1566 – Phishing / T1190 – Exploit Public-Facing Application	Possible entry vectors to discuss during root-cause investigation.	Assumption for facilitation; not specified in STIX.
Credential Access / Defense Evasion	T1078 – Valid Accounts	Potential use of compromised credentials to access file repositories, scheduling services, or administrative shares.	Plausible ransomware path; validate during exercise.
Discovery	T1083 – File and Directory Discovery; T1018 – Remote System Discovery	Adversary identifies shared repositories, production-support servers, and valuable data stores before encryption.	Plausible ransomware path; supports detection discussion.
Lateral Movement	T1021 – Remote Services; T1570 – Lateral Tool Transfer	Ransomware reaches multiple servers and shared repositories.	Plausible ransomware path; not explicitly specified.
Collection / Exfiltration	T1005 – Data from Local System; T1041 – Exfiltration Over C2 Channel	Ransom note threatens disclosure; no confirmed exfiltration in base scenario.	Optional pressure point; do not treat as confirmed unless facilitator adds evidence.
Impact	T1486 – Data Encrypted for Impact	Scheduling systems and shared repositories become unavailable due to ransomware encryption.	Scenario-confirmed by ransomware disruption.
Impact	T1489 – Service Stop	Services supporting scheduling, file access, inventory visibility, or monitoring may be stopped to enable encryption.	Plausible ransomware behavior for discussion.
Impact	T1490 – Inhibit System Recovery	Backup and recovery controls should be checked for deletion, corruption, or tampering.	Plausible ransomware behavior; important recovery control.
Impact	T1499.004 – Endpoint Denial of Service: Application or System Exploitation	Business applications become unavailable, causing operational denial of service.	Scenario impact reflected in scheduling and dashboard outages.

Suggested control themes linked to the mapping

- Harden identity and privileged access; review domain admin use, service accounts, and conditional access.
- Maintain isolated, immutable, and regularly tested backups for production-support systems and quality records.

- Detect mass file modification, ransom note creation, service stop events, shadow copy deletion, and unusual authentication patterns.
- Segment production-support systems from general corporate networks and validate reconnection criteria after containment.
- Predefine business restoration priorities for scheduling, inventory, quality records, and cold chain evidence.

7. Glossary

Term	Definition
BCP	Business Continuity Plan.
Cold chain	Temperature-controlled handling, storage, and transport of perishable products.
Crisis cell	Cross-functional leadership group coordinating business response to a major incident.
IOC	Indicator of Compromise; evidence such as hashes, IPs, domains, process names, or behaviors.
RTO / RPO	Recovery Time Objective and Recovery Point Objective.
SOC	Security Operations Center responsible for cyber monitoring, triage, and technical incident coordination.
STIX	Structured Threat Information Expression, a standard format for cyber threat intelligence.

Source STIX summary

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